

CSE475 — Task 3: Ablation Study and Final Model

NinaPro DB1 EMG Gesture Classification (Reduced Scope)

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Abstract

This report presents Task 3 of a gesture-classification project on NinaPro DB1: a systematic ablation study over eight architectural and training-strategy axes (24 configurations total), followed by 5-fold subject-wise cross-validation of the selected final configuration and a paired Wilcoxon significance test against a same-folds classical baseline. Due to compute-time constraints, this ablation was run on a *reduced data scope* relative to Task 2: Exercise A (E1) only, EMG-only (10 nodes, not 32), 13 classes including rest, rather than Task 2’s EMG+glove, all-exercises, 23-class, rest-excluded setup. This scoping difference is made explicit throughout, and the final model’s Macro-F1 = 0.1433 ± 0.0142 is compared honestly against a paired Random Forest baseline (Macro-F1 = 0.2238 ± 0.0369), showing the GNN underperforms even after systematic tuning, with the difference not reaching statistical significance at $n = 5$ folds (Wilcoxon $p = 0.0625$).

1 Scope and Data Setup

1.1 Explicit Scoping Difference from Task 2

For compute-time reasons, an 8-axis architecture search with 5-fold cross-validation was not tractable on the full Task 2 data scope (EMG+glove, all 3 exercises, 23 classes) within the available GPU budget. Task 3 therefore uses a deliberately reduced scope:

Table 1: Data scope comparison: Task 2 (proposed GNN) vs. Task 3 (ablation).

	Task 2 (Proposed GNN)	Task 3 (Ablation)
Exercises used	All 3 (A, B, C)	Exercise A (E1) only
Signal channels (graph nodes)	32 (10 EMG + 22 glove)	10 (EMG only)
Classes	23 (rest excluded)	13 (rest <i>included</i> as class 0)
Node feature descriptors	RMS, MAV, WL, VAR (4)	MAV, RMS, WL, ZC, SSC (5)
Windowed samples	30,936	133,486

This is a legitimate, deliberate scoping decision, not an inconsistency to be hidden: the ablation sweep runs 24 separate training configurations, and a 5-fold cross-validation of the winning configuration on top of that, which was only feasible within the assignment’s time budget on the smaller, single-exercise, EMG-only scope. Task 2 and Task 3 results are therefore *not* directly comparable to one another and are reported as two separate lines of evidence rather than combined into a single table.

1.2 Data Loading and Windowing

Data is loaded from all 81 raw `.mat` files ($27 \text{ subjects} \times 3 \text{ exercises}$), each containing `emg` (10 channels), `restimulus`, `repetition/rerepetition`, `subject`, and `exercise` fields at 100 Hz. Only the 27 files matching Exercise A (E1) are used for Task 3. Signals are windowed (200-sample window, 100-sample step) into node feature vectors of five Hudgins time-domain descriptors (MAV, RMS, WL, ZC, SSC) per channel, giving a $133,486 \times 54$ windowed table (50 features + label + subject + repetition + exercise).

Table 2: Class distribution (Exercise A, rest included as class 0).

Class	0 (rest)	1	2	3	4	5	6
Count	75,590	5,008	4,950	5,560	4,620	4,634	4,840
Class	7	8	9	10	11	12	
Count	4,898	5,077	4,669	4,491	4,309	4,840	

Because rest is retained as a class in Task 3 (unlike Task 1/2, where it was excluded), the imbalance ratio here is far more severe than reported in the Task 1 EDA: $\text{max/min class count}$ is $75,590/4,309 = 17.5\times$, driven entirely by the dominant rest class. This has a direct, visible effect on one ablation result discussed in Section 3.

1.3 Node Feature Normalization

Node features (MAV, RMS, WL, ZC, SSC) span very different numeric scales (RMS is a small voltage-like value; ZC/SSC are integer counts up to ~ 20). All features are standardized (fit on the training split only) before graph construction, inside the same function call used for every ablation experiment and every cross-validation fold, ensuring the fit is always leakage-safe and never reuses a scaler across splits.

2 Ablation Methodology

2.1 Graph Variants

Two graph constructions are compared (Ablation 4): a **correlation graph** (edges from training-set channel correlation, thresholded, with a k -NN fallback if too sparse) and a fixed **anatomical adjacency graph** (the 8 equally-spaced forearm electrodes wired as a ring, with the 2 extra muscle-specific electrodes each linked to two ring electrodes).

2.2 Architectures Compared

Three graph convolution operators are compared (Ablation 5): GCN, GAT (GATv2), and GraphSAGE, via a single `ConfigurableGNN` class parameterized by architecture, hidden dimension, number of layers, dropout, and pooling strategy.

2.3 The Eight Ablation Axes

A1 Number of GNN layers: 1, 2, 3

A2 Hidden dimension: 32, 64, 128

A3 Dropout: 0.0, 0.2, 0.5

A4 Graph construction: fixed adjacency vs. correlation

A5 Architecture: GCN, GAT, GraphSAGE

A6 Edge weighting: unweighted vs. weighted

A7 Pooling: mean, max, mean+max concatenation

A8 Training strategy: Adam vs. AdamW; scheduler on/off; weighted vs. unweighted loss

Each of the 24 resulting configurations (1 baseline + 23 single-axis variants) is trained in *fast mode* (15 epochs max, patience 5) on one held-out subject-wise split (79,162 train / 24,704 validation / 29,620 test windows) so that all experiments are directly comparable against a common baseline.

3 Ablation Results

Table 3: Full ablation sweep results (fast mode, 15 epochs), sorted by test Macro-F1. Baseline GNN Macro-F1 = 0.1293 (highlighted); configurations below it are shaded.

Experiment	Configuration	Accuracy	Macro-F1	Weighted-F1
A3_dropout_0.0	{dropout: 0.0}	0.4421	0.1416	0.4612
A4_graph_adjacency	{graph_type: adjacency}	0.4784	0.1354	0.4748
A6_edge_weighted	{edge_weighted: true}	0.4945	0.1330	0.4751
A7_pool_mean	{pooling: mean}	0.5037	0.1313	0.4815
A7_pool_mean_max	{pooling: mean_max}	0.4857	0.1300	0.4741
A1_layers_1	{num_layers: 1}	0.4893	0.1298	0.4807
Baseline GNN	{}	<i>0.4818</i>	<i>0.1293</i>	<i>0.4699</i>
A8_no_scheduler	{use_scheduler: false}	0.4969	0.1282	0.4767
A2_hidden_128	{hidden_dim: 128}	0.4642	0.1278	0.4634
A5_arch_SAGE	{arch: SAGE}	0.4623	0.1275	0.4713
A4_graph_correlation	{graph_type: correlation}	0.4879	0.1274	0.4702
A2_hidden_64	{hidden_dim: 64}	0.4838	0.1264	0.4725
A3_dropout_0.2	{dropout: 0.2}	0.4770	0.1263	0.4678
A7_pool_max	{pooling: max}	0.4950	0.1262	0.4738
A5_arch_GCN	{arch: GCN}	0.4913	0.1257	0.4735
A6_edge_unweighted	{edge_weighted: false}	0.4848	0.1255	0.4707
A1_layers_3	{num_layers: 3}	0.4705	0.1230	0.4669
A2_hidden_32	{hidden_dim: 32}	0.4940	0.1211	0.4706
A3_dropout_0.5	{dropout: 0.5}	0.4971	0.1207	0.4734
A5_arch_GAT	{arch: GAT}	0.4924	0.1206	0.4696
A1_layers_2	{num_layers: 2}	0.4914	0.1205	0.4727
A8_optim_Adam	{optimizer: Adam}	0.4685	0.1204	0.4605
A8_optim_AdamW	{optimizer: AdamW}	0.4718	0.1185	0.4643
A8_unweighted_loss	{weighted_loss: false}	0.5632	<i>0.1015</i>	0.4463

Notable Result: A8_unweighted_loss

Removing class-weighted loss produces the *highest raw accuracy* of all 24 experiments (0.5632) but the *lowest Macro-F1* (0.1015). This is a direct, concrete illustration of the class-imbalance effect flagged in Task 1: with rest (class 0) comprising 56.6% of all windows, an unweighted model can achieve high accuracy simply by favoring the dominant class, at the cost of essentially ignoring minority gesture classes — exactly the metric-choice justification given throughout this project for reporting Macro-F1 rather than accuracy alone.

4 Final Configuration Selection

4.1 Selection Rule and an Important Nuance

The final configuration is assembled by examining each of the eight ablation axes independently: for each axis, if its best-performing variant exceeds the un-ablated Baseline GNN’s Macro-F1 (0.1293), that variant’s setting is adopted; otherwise the axis retains its original default value (from the baseline configuration, which itself defaults to architecture = GAT).

This produces an important, honest subtlety worth stating explicitly: the final selected architecture is GAT — *not because GAT outperformed GCN or GraphSAGE in the ablation*. In fact, Table 3 shows **A5_arch_GAT** (Macro-F1 = 0.1206) scored *below* all three of GraphSAGE (0.1275), GCN (0.1257), and even the baseline itself (0.1293). GAT remained the final architecture only because it was already the pipeline’s default and no architecture variant in the A5 sweep exceeded the baseline threshold required to trigger a change. This should be reported plainly: the ablation study demonstrates that, in this fast-mode 15-epoch setting, no architecture choice measurably improved over the baseline, which is itself a valid and informative negative finding about this axis.

Table 4: Per-axis outcome: which variant was adopted and why.

Axis	Best variant found	Beat baseline?	Adopted
A1 (layers)	1 layer (0.1298)	Yes (barely)	num_layers = 1
A2 (hidden dim)	128 (0.1278)	No	hidden_dim = 64 (unchanged default)
A3 (dropout)	0.0 (0.1416)	Yes	dropout = 0.0
A4 (graph)	adjacency (0.1354)	Yes	graph_type = adjacency
A5 (architecture)	SAGE (0.1275)	No	arch = GAT (unchanged default)
A6 (edge weight)	weighted (0.1330)	Yes	edge_weighted = true
A7 (pooling)	mean (0.1313)	Yes	pooling = mean
A8 (training)	no-scheduler (0.1282)	No	optimizer/scheduler/loss unchanged

Table 5: Final selected configuration.

Parameter	Value
Architecture	GAT (GATv2), retained as default (see nuance above)
Hidden dimension	64
Number of layers	1
Dropout	0.0
Pooling	mean
Edge weighting	weighted
Graph construction	fixed anatomical adjacency
Optimizer	AdamW
LR scheduler	On (ReduceLROnPlateau)
Loss weighting	Class-weighted
Training epochs (final fit)	80 (patience 12)

5 Final Model: 5-Fold Subject-Wise Cross-Validation

The selected configuration is retrained in full (80 epochs, patience 12) across 5 subject-wise `GroupKFold` folds (DB1’s 27 subjects comfortably support a valid 5-fold split). A Random Forest classifier, re-trained on the identical fold splits using the same 50-dimensional feature table, serves as a paired baseline for a fair, same-folds comparison.

Table 6: Per-fold results: Final GNN vs. paired Random Forest baseline.

Fold	Train/Val/Test size	GNN Macro-F1	RF Macro-F1	Diff
1	89,013 / 19,740 / 24,733	0.1632	0.2229	−0.0597
2	88,975 / 19,780 / 24,731	0.1305	0.1622	−0.0317
3	88,910 / 19,844 / 24,732	0.1295	0.2297	−0.1003
4	84,056 / 19,785 / 29,645	0.1507	0.2504	−0.0997
5	84,132 / 19,709 / 29,645	0.1425	0.2540	−0.1115

Table 7: 5-fold cross-validation summary (mean $\pm$ std).

Metric	Final GNN	Paired Random Forest
Accuracy	0.4079 $\pm$ 0.0428	0.5978 $\pm$ 0.0364
Macro-F1	0.1433 $\pm$ 0.0142	0.2238 $\pm$ 0.0369
Weighted-F1	0.4503 $\pm$ 0.0342	—
Precision (macro)	0.1527 $\pm$ 0.0123	—
Recall (macro)	0.1576 $\pm$ 0.0176	—

In every one of the 5 folds, the Random Forest baseline outperforms the final GNN on Macro-F1, with gaps ranging from -0.032 to -0.112 .

6 Statistical Significance: Wilcoxon Signed-Rank Test

A paired Wilcoxon signed-rank test compares the final GNN’s per-fold Macro-F1 against the Random Forest baseline’s per-fold Macro-F1 (Table 6).

Table 8: Wilcoxon signed-rank test result.

Test statistic	0.0000
p -value	0.0625
α	0.05
Statistically significant at $\alpha = 0.05$?	No

With only $n = 5$ paired folds, the test is underpowered: 5 paired differences is the minimum sample size at which the Wilcoxon test can ever report $p < 0.05$ (achieved only if all 5 differences share the same sign, which is exactly what occurs here — the GNN loses on all 5 folds, yet $p = 0.0625$ remains just above the 0.05 threshold). This should be reported honestly as a *consistent but not formally significant* disadvantage, rather than either overstating significance or dismissing the consistent directional pattern across all 5 folds.

7 Final Comparison and Discussion

Table 9: Final Task 3 comparison table.

Model	Accuracy	Macro-F1
Best Baseline (Random Forest, paired fold estimate)	0.5978	0.2238
Final Improved GNN (5-fold CV mean)	0.4079	0.1433

Honest Reading of Results

1. **The ablation study did not close the gap with classical baselines.** Even after systematically searching 8 architectural/training axes, the best achievable fast-mode Macro-F1 (0.1416, from A3_dropout_0.0) remained far below the paired Random Forest’s fully-trained 5-fold mean of 0.2238.
2. **The architecture axis (A5) is the study’s most informative negative result.** No graph convolution variant (GCN, GAT, GraphSAGE) improved on the baseline in this setting, meaning the final model’s ”GAT” label reflects an unchanged default rather than an empirically validated choice — an important distinction to state plainly in any report or viva defense of this work.
3. **Severe rest-class imbalance ($17.5\times$) dominates this scope’s difficulty**, as concretely demonstrated by the A8_unweighted_loss result (accuracy 0.563, Macro-F1 0.102): a model can trivially inflate accuracy by favoring rest while providing almost no discriminative value for actual gestures.
4. **The Wilcoxon result is directionally consistent but not formally significant** at $n = 5$ folds — the GNN underperforms on every single fold, but 5 folds is the practical floor for this test to ever reach significance, and this limitation is reported explicitly rather than glossed over.
5. **Scope difference from Task 2 remains a caveat.** Because Task 3 uses a reduced scope (Exercise A only, EMG-only, rest included), these results cannot be combined with Task 2’s EMG+glove/23-class findings into a single ”final” number; both are reported as separate, valid pieces of evidence about this GNN-based approach’s performance relative to classical baselines.

8 Conclusion

A systematic 8-axis, 24-configuration ablation study followed by rigorous 5-fold subject-wise cross-validation and paired significance testing was carried out on a reduced-scope (Exercise A, EMG-only, 13-class) version of NinaPro DB1. The resulting best configuration (GAT architecture retained by default, 1 layer, no dropout, mean pooling, weighted anatomical-adjacency graph) achieves $\text{Macro-F1} = 0.1433 \pm 0.0142$, underperforming a paired Random Forest baseline ($\text{Macro-F1} = 0.2238 \pm 0.0369$) on every fold, though the Wilcoxon test does not reach formal significance at this fold count ($p = 0.0625$). The most actionable finding from the ablation itself is that no architecture variant tested measurably improved over the baseline GNN in fast-mode training, suggesting that, within this project’s compute budget and data scope, architectural tuning alone

is insufficient to close the gap with classical methods; addressing the severe class imbalance and revisiting the feature/graph representation are identified as more promising directions for future work than further architecture search.